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A Real-time Path Planner based on Improved Tentacle Algorithm for Autonomous Vehicles

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***Abstract*—In this paper, a real-time local path planner for obstacle avoidance of autonomous vehicles is proposed. Based on the tentacle algorithm, a novel tentacle curve that can ensure the continuity of the derivative of the path curvature is used to generate an alternate path set. After collision detection, we use the analytic hierarchy process (AHP) to calculate the weights to choose the best path. Finally, to verify the efficiency of the proposed planner, an obstacle avoidance case is designed in the simulation. The simulation results show that the proposed planner can obviously avoid the obstacle well and has better smoothness than the tentacle approach that uses the clothoid curve. The total time of local planning within a single period is less than 20 ms, which still has great real-time performance.**

Keywords-autonomous vehicle; path planning; improved tentacle algorithm; obstacle avoidance

I. INTRODUCTION

Autonomous vehicles are important means for alleviating traffic pressure, strengthening road safety and reducing energy consumption; so, they have attracted much attention. Motion planning is one of the core areas of intelligent vehicle implementation and it has always been at the center of academic and industrial research [1].

The graph search algorithm is a traditional planning algorithm. This kind of algorithm discretizes the state space on a graph and then searches for a path with the lowest cost function on the graph. The Dijkstra algorithm [2] is a typical graph search algorithm. Afterward, the A-star algorithm was proposed, to reduce the search time by introducing a heuristic function [3]. The Hybrid A-star algorithm, which considers the vehicle kinematic constraint model, was proposed later [4]. The sampling-based algorithm is a commonly used planning algorithm. A typical sampling-based algorithm is RRT [5], which starts from the root node and spreads around to obtain a feasible path. However, the path generated by RRT is not continuous and smooth, which is bad for vehicle tracking. The second type of sampling-based algorithm is the state space sampling algorithm, in which, the sampling results are combined with numerical optimization [6] or parametric curves [7] to generate a path set. However, this algorithm requires many calculations. Another typical algorithm is the tentacle algorithm, which tentatively tries multiple directional control schemes to establish candidate paths. Since the sampling of tentacles is only related to the current state of the vehicle, its calculation time is relatively small. In [8], a circular arc is used to construct tentacles for path planning. The optimal path is selected according to the obstacle location, reference path and road smoothness. In [9], a clothoid curve replaces the circular arc to generate the tentacle and ensure the continuity of the path curvature. In addition, trajectory planning combining the tentacle sampling method with a node search algorithm is proposed in [10].

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To ensure that the steering angle change rate is continuous when the vehicle tracks a trajectory, this paper proposes a real-time obstacle avoidance local trajectory planner, based on the tentacle algorithm. We propose a curve with a continuous derivative of curvature to construct the tentacles, to generate the paths to be selected. After collision detection, the optimal candidate path is selected by using the analytic hierarchy process (AHP) to design the preferred weights. Finally, we implement an obstacle avoidance scenario through a simulation to verify the planner. The results show that our algorithm avoids obstacles well and the rate of change in the steering wheel angle is sufficiently smooth. The contributions of this work are as follows:

- A time-efficient path planning method based on a tentacle curve with continuous derivative is proposed.
- A path selection weight design method based on the analytic hierarchy process is proposed.

The remainder of this paper is structured as follows. In Section II, we present the path planning algorithm based on tentacles with curvatures having continuous derivatives. In Section III, we explain how we use AHP to select the optimal path. Simulation results and an analysis of obstacle avoidance are discussed in Section IV, followed by the conclusion in Section V.

II. LOCAL PATH PLANNER

In this section, we use a curve with a curvature that ensures the continuity of the derivatives to build the tentacle path set. The vehicle kinematics model is presented as in (1):

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\phi} \end{bmatrix} = v \begin{bmatrix} \cos\phi \\ \sin\phi \\ \dfrac{\tan\delta}{L} \end{bmatrix} \tag{1}$$

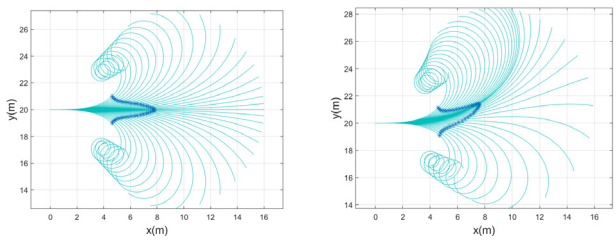
where is the vehicle heading angle and is the steering angle.

To ensure the continuity of the curvature derivatives of locally planned paths, the curvature derivatives of tentacles are defined as:

$$\frac{d\rho}{ds} = \beta s^{0.5} + C_0 \tag{2}$$

where, s is the curve length and $\in$ [-0.055,0.055] represents the change speed of the curvature derivative. The curve length C_0 is the derivative of the path curvature when s=0, which is the initial curvature derivative relative to the current steering angle.

Taking an indefinite integral of both sides of (2), we can obtain the curvature of tentacles as (3),



(a) (b)

Figure 1. Curves at different initial steering angles: (a) C_0 = 0, (b) C_0 = 0.02

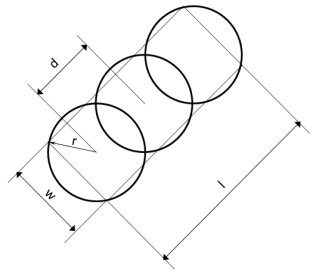


Figure 2. Collision detection configuration

$$\rho(s) = \frac{\beta}{1.5} s^{1.5} + C_0 s + \rho_0 \tag{3}$$

where is the initial curvature of the curve.

Then, the final tentacle candidate path coordinates are presented specific to (4),

$$\begin{cases} x(s) = \int_0^s \cos\left(\dfrac{\beta}{3.75} s^{2.5} + \dfrac{C_0}{m+1} s^{1.5} + \rho_0 s\right) d\tau \\ y(s) = \int_0^s \sin\left(\dfrac{\beta}{3.75} s^{2.5} + \dfrac{C_0}{m+1} s^{1.5} + \rho_0 s\right) d\tau, \quad s \in [0, L] \\ \phi(s) = \int \rho(s) = \dfrac{\beta}{3.75} \end{cases} \tag{4}$$

Considering that when the vehicle speed increases, the detection distance should also increase accordingly, to be able to coordinate with the vehicle speed, the change in the detection length of the tentacle is defined as in (5),

$$l_{ck} = \frac{(l_{ck\max} - l_{ck\min})}{2} \times \sin\left[\frac{\pi}{20} \times (k-11)\right] + \frac{(l_{ck\max} + l_{ck\min})}{2} \tag{5}$$

where l_{ck} is the detection length of the k th tentacle, l_{ckmax} is the detection length of the middle tentacle, and l_{ckmin} is the detection length of the outermost tentacle. The curves computed for different initial steering angles are shown in Fig.1.

III. COLLISION DETECTION AND PATH SELECTION

After generating the tentacle set, the path selector will pick out the best path. The path selector consists of the collision detection and the path selection. Collision detection is used to ensure the safety of the path. To calculate the grid occupied by the vehicle configuration, the vehicle is divided into several circles, as shown in Fig. 2. If one of the circles is unsafe, the

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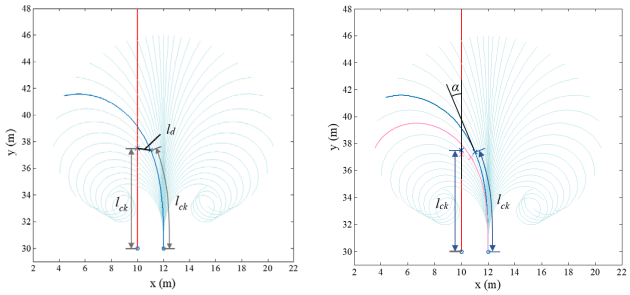
path is marked to be unsafe. All unsafe paths from candidate paths will be eliminated.



(a) (b)

Figure 3. Definition of selection criteria

After collision detection, it is necessary to select an optimal tentacle among the feasible tentacles for the vehicle path to be executed. Only one tentacle among navigable tentacles is chosen as the best tentacle using three cost values, V_{ld}, V_α and V_C. The total cost value of the tentacles is specific as in (6):

$$V = \omega_1 \cdot V_{ld} + \omega_2 \cdot V_\alpha + \omega_3 \cdot V_C \tag{6}$$

where V_{ld} represents the distance criteria, which is defined in (7). V is defined in (8), which represents the angle criteria, and V_C is defined in (9), which represents the curvature criteria. $_1$, $_2$ and $_3$ are the weights of each criterion, which satisfy $_1$+ $_1$+ $_3$ = 1. The total cost value of each tentacle is calculated, and the tentacle with the smallest cost is chosen as the execution path.

$$V_{ld} = \frac{l_d - \min(l_d)}{\max(l_d) - \min(l_d)} \tag{7}$$

$$V_\alpha = \frac{\alpha_k - \min(\alpha_k)}{\max(\alpha_k) - \min(\alpha_k)} \tag{8}$$

$$V_C = \frac{|C_k| - \min(|C_k|)}{\max(|C_k|) - \min(|C_k|)} \tag{9}$$

As shown in Fig. 3(a), l_d is defined as the deviation distance of the kth tentacle path from the reference path at length l_{ck}; min(l_d) is the minimum l_d of all tentacle paths, and max(l_d) is the maximum. As shown in Fig. 3(b), α_k is defined as the included angle between the tangent direction of the kth tentacle and the reference path at l_{ck}. The min(α_k) is the minimum α_k of all tentacle paths, and max(α_k) is the maximum. C_k is the derivative of the curvature of the kth tentacle, and max($|C_k|$)/min($|C_k|$) is the maximum/minimum $|C_k|$ of all tentacles.

The analytic hierarchy process decomposes complex problems into several levels for analysis. Here, it is used to determine the coefficients in path selection. In this paper, we set accuracy criterion (10) and smoothness criterion (11) as evaluation criteria.

$$\begin{cases} e_{\max} = \max(e_i) \\ \bar{e} = \dfrac{1}{M}\sum_{i=1}^{M} e_i \end{cases} \tag{10}$$

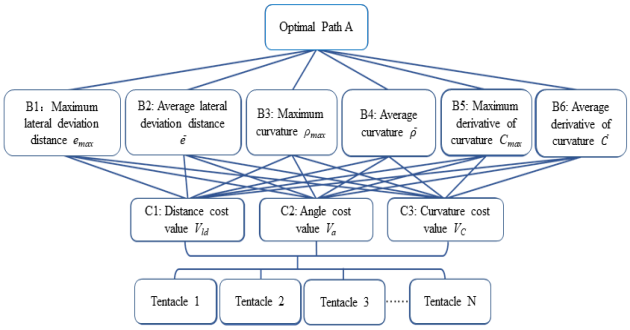


Figure 4. AHP path selection model

$$\begin{cases} \rho_{\max} = \max(|\rho_i|) \\ \bar{\rho} = \dfrac{1}{M}\sum_{i=1}^{M}|\rho_i| \\ C_{\max} = \max(|C_i|) = \max\left(\left|\dfrac{d\rho_i}{ds}\right|\right) \\ \bar{C} = \dfrac{1}{M}\sum_{i=1}^{M}|C_i| = \dfrac{1}{M}\sum_{i=1}^{M}\left|\dfrac{d\rho_i}{ds}\right| \end{cases} \tag{11}$$

where e_i is the lateral deviation distance between each path point on the tentacle path and the corresponding point on the reference path. The e_{max} is the maximum lateral deviation distance, and the $\bar{e}$ is the average lateral deviation distance. ρ_i is the curvature between each path point on the tentacle path and the corresponding point on the reference path. ρ_{max} is the maximum curvature of the planning path, and $\bar{\rho}$ is the average curvature. C_i is the curvature derivative between each path point on the tentacle path and the corresponding point on the reference path. C_{max} is the maximum curvature derivative of the planning path, and $\bar{C}$ is the average curvature derivative.

Based on these costs and criteria, the AHP path selection model operates as shown in Fig. 4. Different layers represent different hierarchies, and the lines indicate relations.

It is assumed that an element A_i is connected to lower layer elements $B_i \ldots B_N$, where the judgment matrix $\boldsymbol{B}$ is as shown in Table I.

TABLE I. JUDGMENT MATRIX FOR A_i

The lower element of A_i	B_1	B_2	…	B_N
B_1	b_{11}	b_{12}	…	b_{1N}
B_2	b_{21}	b_{22}	…	b_{2N}
…	…	…	…	…
B_N	b_{N1}	b_{N2}	…	b_{NN}

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The elements $b_{ij} \in [0, 9]$ indicates the importance of B_i relative to B_j for A_i. The larger b_{ij} is, the more important B_i is relative to B_j. In addition, $b_{ii}= 1$ and $b_{ji}= 1/b_{ij}$.

To ensure the consistency of the judgment matrix, a consistency test is needed. The consistency ratio is defined as (12)

$$CR = \frac{CI}{RI} \tag{12}$$

where $CI = (\lambda_{max}-N)/(N-1)$ is the consistency indicator of the matrix, λ_{max} is the maximum eigenvalue of the judgment matrix, N is the dimension of the judgment matrix and RI is the corresponding average stochastic consistency indicator. When CR<0.1, the consistency of the matrix is acceptable.After calculating the weights between the elements and testing their consistency, the weights for our AHP model can be obtained. We define the weight between elements C_i and B_j as ω_{Ci_Bj} and the weight between B_j and A as ω_{Bj}. Then, the total weight between C_i and A is defined as (13).

TABLE II. THE JUDGMENT MATRIX FOR B_i

The lower element of B_1	C_1	C_2	C_2	$\omega_{C_i_B_1}$
C_1	1	9	4	0.7375
C_1	1/9	1	1/2	0.0852
C_1	1/4	2	1	0.1733
The lower element of B_2	C_1	C_2	C_2	$\omega_{C_i_B_2}$
C_1	1	9	4	0.7267
C_1	1/9	1	1/3	0.0734
C_1	1/4	3	1	0.2000
The lower element of B_3	C_1	C_2	C_2	$\omega_{C_i_B_3}$
C_1	1	1/4	1/2	0.1365
C_1	4	1	3	0.6250
C_1	2	1/3	1	0.2385
The lower element of B_4	C_1	C_2	C_2	$\omega_{C_i_B_4}$
C_1	1	1/2	3	0.3196
C_1	2	1	4	0.5584
C_1	1/3	4	1	0.1220
The lower element of B_5	C_1	C_2	C_2	$\omega_{C_i_B_1}$
C_1	1	1/4	1/4	0.1111
C_1	4	1	1	0.4444
C_1	4	1	1	0.4444
The lower element of B_6	C_1	C_2	C_2	$\omega_{C_i_B_6}$
C_1	1	9	4	0.7375
The lower element of B_1	C_1	C_2	C_2	$\omega_{C_i_B_1}$
C_1	1/9	1	1/2	0.0852
C_1	1/4	2	1	0.1733

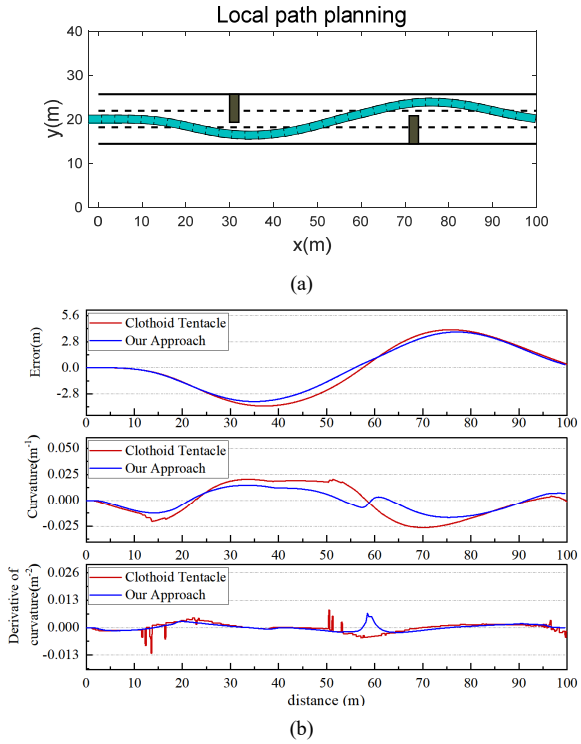

Figure 5. Simulation result: (a) Local path planning result, (b) Comparison chart of evaluation indicators with clothoids

$$\omega_{C_i} = \sum_{j}^{m} \omega_{C_i_B_j} \cdot \omega_{B_j} \tag{13}$$

where m is the number of B-layer elements. Finally, the test equation for the consistency of the total judgment matrix is shown in (14).

$$CR = \frac{CI}{RI} = \frac{\sum_{j=1}^{m} CI_{B_j} \cdot \omega_{B_j}}{\sum_{j=1}^{m} RI_{B_j} \cdot \omega_{B_j}} \tag{14}$$

where CI_{B_j} and RI_{B_j} are the consistency indicators and the average random consistency indicators of the judgment matrix, which consist of the lower elements of layer B.

In the obstacle avoidance scenario, the maximum lateral deviation is mainly influenced by the distance cost V_{ld}. The second influencer is the angle cost V_α and the last is the curvature cost V_C. Therefore, the order of importance is $V_{ld} > V_C > V_\alpha$, and we can obtain each judgment matrix from elements between layer B and layer C, which are provided in Table II.

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The result of the judgment matrix calculation is substituted into Eq. (14), to obtain the weight of each cost under the obstacle avoidance simulation. In the end, the weights of each criterion are found as ω_1 = 0.4954, ω_2 = 0.2716, and ω_3 = 0.2330.

IV. Simulation Results and Analysis

To verify the planning effect of the proposed curve, we built a simulation scenario in MATLAB, in which the tentacle sampling time T= 0.1 s. Then we compared the results with the clothoid algorithm. The simulation results are shown in Fig. 5.

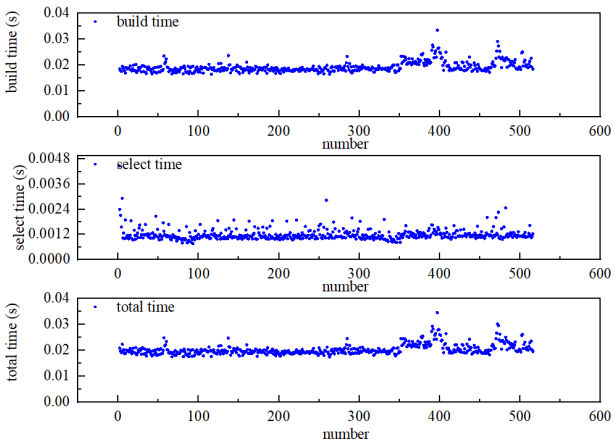

Figure 6. Time spent in local path planning

As shown in Fig. 5(a), the simulation scenario consists of a one-way, three-lane road and two obstacles. The obstacles are located on both sides of the road, and their longitudinal distances from the starting point are 31 meters and 72 meters, respectively. The autonomous vehicle successfully completes the local path planning for obstacle avoidance. Meanwhile, in Fig. 5(b), we compare our approach and clothoid tentacle from three aspects: the lateral error, curvature and change rate of curvature.

Some specific evaluation indicators of the obstacle avoidance simulation are listed in Table III. With our approach, the average curvature and average curvature derivative were both reduced to half of the clothoid tentacle, and the maximum curvature and maximum curvature derivative were far below those of clothoid.

TABLE III. Evaluation Indicators

	Maximum		Average	
	$\lvert \rho_{max} \rvert$ (m^{-1})	$\lvert d\rho \rvert_{max}$ (m^{-2})	$\bar{\rho}$ (m^{-1})	$\overline{d\rho}$ (m^{-2})
Clothoid tentacle	0.0263	0.0121	0.01361	0.00156
Our method	0.0167	0.0068	0.00891	0.00119

Figure 6 shows the computation time for tentacle generation and selection with the proposed curve in 500 tests of our planning algorithm.

In 500 cycles of local path planning simulation, the average time for generating tentacles was 19.3 ms, and the probability within 20 ms was 77.79%. The average time for dividing and selecting tentacles was 1.2 ms, and the probability within 2 ms was 97.68%. The total time for local planning was 20.57 ms, and the probability within 25 ms was 95.36%.

It can be seen from the total planning time that the algorithm has good real-time performance and can basically guarantee the completion of local path planning within 25 ms.

V. Conclusion

A local path planner aims to generate a safe and smooth path for autonomous vehicles. To solve the problem regarding the discontinuity of the curvature differential of the path, this paper proposes a new curve based on the tentacles, which is used to generate candidate routes. Based on the analytic hierarchy process (AHP), we design the preferred weights to select the best candidate path. Simulation results show that our work has better smoothness compared to the clothoid tentacle algorithm. After 500 tests in the obstacle avoidance scenario, it is found that the proposed curve can ensure the completion of the generation, division and updating of tentacles in a short time and also has good real-time performance.

Future work will focus on dynamic obstacle avoidance by combining the tentacle algorithm with optimization methods.

Acknowledgment

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